

PINAR YILMAZ

Electrical-Electronics Engineer (New Graduate)

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PROFESSIONAL SUMMARY

Recent Electrical-Electronics Engineering graduate with hands-on project experience in embedded systems, PCB design, and AI-based image processing. Completed three internship rotations building hardware-software integration on STM32 and ESP32 platforms, including communication protocols (MQTT, UART, GSM) and PCB development. Active contributor to a TÜBİTAK 2209-A funded smart-glasses project for visually impaired users.

WORK EXPERIENCE

MASSWELL — Engineering Intern, PCB Design

Apr 2026 – May 2026

Technologies: Altium Designer, PCB Design, STM32, Soldering

- Led PCB design for an STM32-based smart security system, from schematic design through manufacturing files.
- Prepared PCB routing and production (Gerber) files for fabrication.
- Performed board assembly, soldering, and hardware validation testing.

MASSWELL — Engineering Intern, Embedded Systems

Feb 2026 – Mar 2026

Technologies: ESP32, C++, Arduino IDE, SD Card Module, MQTT

- Built a system that collects battery data over UART via ESP32 and logs it to an SD card with RTC integration.
- Designed a desktop application enabling real-time monitoring and file download without physically removing the SD card, reducing field maintenance time.

MASSWELL — Engineering Intern, Communication Systems

Jul 2025 – Aug 2025

Technologies: ESP32, C++, Arduino IDE, GSM Module, MQTT

- Implemented hardware-software integration for battery management system communication via the SIM800L module.
- Built remote monitoring infrastructure using MQTT over TCP for real-time data transmission.

PROJECTS

3D Audio & AI-Assisted Obstacle Detection Glasses — TÜBİTAK 2209-A

2025 – 2026

Technologies: YOLO, Raspberry Pi, CNN/RNN, 3D Audio

- Co-developing smart glasses providing real-time object detection and directional audio feedback for visually impaired users.
- Integrated dual-camera hardware, embedded systems, and image processing pipelines.

Breast Cancer Classification (Deep Learning)

Apr 2025 – May 2025

Technologies: MATLAB, Jupyter, Python | Report & GitHub available

- Classified histopathological breast cancer images using InceptionResNetV2 and VGG16 deep learning models, benchmarked against traditional machine learning methods.
- Optimized image preprocessing and segmentation, achieving 90% classification accuracy.

Line-Following Robot with Adaptive PID Speed Control

Nov 2024 – Dec 2024

Technologies: C, STM32, PID | Report available

- Developed an STM32-based line-following robot using PID control to minimize line deviation and improve driving stability.
- Fine-tuned motor speed and direction via PWM signals for sharp turns and smooth line tracking.

Digital Stethoscope

Nov 2023 – Jan 2024

Technologies: MATLAB, PCB, Arduino

- Designed a low-cost electronic stethoscope using digital and analog filters for clean heart-signal acquisition.

EDUCATION

Abdullah Gül University

2021 – 2026

B.Sc., Electrical-Electronics Engineering (100% English) — GPA: 3.13 / 4.00

TECHNICAL SKILLS

Programming: Python, C++, C, MATLAB

Embedded Platforms: STM32, ESP32, Raspberry Pi, Arduino

Communication Protocols: UART, MQTT, GSM/SIM800L, I2C, SPI

Signal Processing: FFT, Filtering, Time-Frequency Analysis

Software/Tools: Altium Designer, Simulink, PSpice, LTspice, OpenCV, NumPy, SciPy, Matplotlib

Languages: Turkish (Native), English (Advanced)